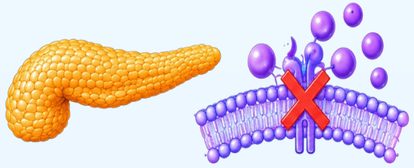
图 1 糖尿病肾病及其心血管并发症的共同病理机制

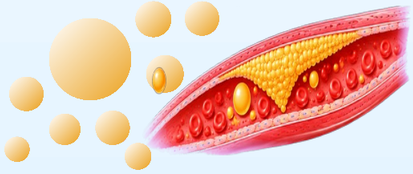
1 代谢异常



高血糖

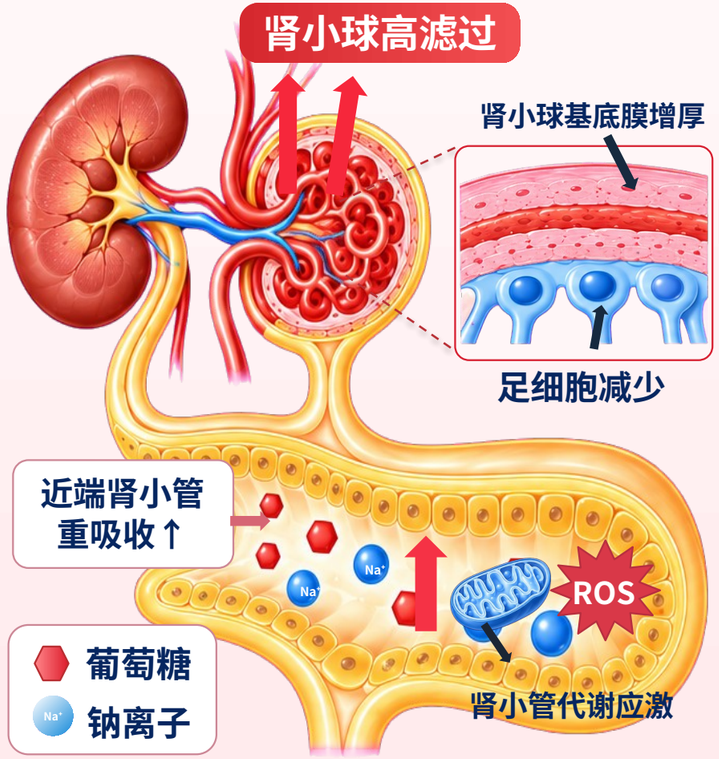


胰岛素抵抗



脂质代谢异常

2 肾脏早期损伤

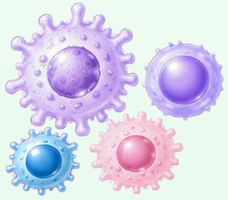


3 炎症 - 纤维化轴

RAS 激活

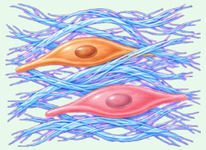


氧化应激



炎症细胞募集

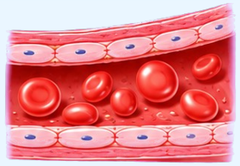
巨噬细胞、 T 淋巴细胞等



胶原沉积 / 纤维化

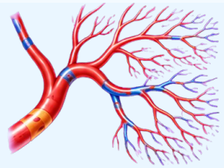
成纤维细胞活化、 ECM 沉积

4 血管与心脏损伤



内皮功能障碍

NO 减少、炎症因子增加



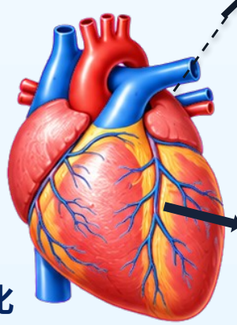
微循环异常

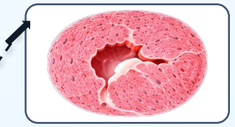
毛细血管稀疏、灌注不足



动脉粥样硬化

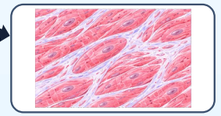
脂质沉积、斑块形成





左心室肥厚

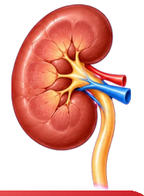
心肌重构



心肌纤维化

心肌僵硬、舒张功能受损

心力衰竭风险↑



肾脏损伤

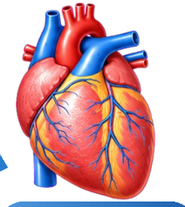
肾功能下降加重心脏负荷

（容量负荷、毒素蓄积、 RAS 激活等）

心肾相互促进，形成恶性循环

心功能障碍加重肾脏损伤

（肾灌注减少、静脉淤血、神经体液激活等）



心脏损伤

共同病理基础：代谢负荷增加 + 血流动力学紊乱 + 炎症与纤维化 + 血管内皮损伤